

Math 135 Homework 7
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Problem 1

Find the stationary function of

$$J[y] = \int_0^1 [(y')^2 + y^2] dx, \quad y(0) = 0, \quad y(1) = 1.$$

The integrand is

$$F = (y')^2 + y^2.$$

Euler Lagrange:

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \frac{\partial F}{\partial y'} = 0.$$

Compute:

$$F_y = 2y, \quad F_{y'} = 2y'.$$

Then:

$$2y - \frac{d}{dx}(2y') = 0.$$

Divide by 2:

$$y - y'' = 0.$$

Solve:

$$y'' - y = 0, \quad y(x) = Ae^x + Be^{-x}.$$

Apply conditions:

1. $y(0) = 0$:

$$A + B = 0 \Rightarrow B = -A.$$

2. $y(1) = 1$:

$$A(e - e^{-1}) = 1.$$

Thus:

$$A = \frac{1}{e - e^{-1}}, \quad B = -A.$$

Final answer:

$$y(x) = \frac{e^x - e^{-x}}{e - e^{-1}}.$$

Problem 2

Minimize

$$J[y] = \int_0^1 [(y')^2 + y^2] dx, \quad y(0) = 0, \quad y(1) \text{ free.}$$

Euler Lagrange gives the same ODE:

$$y'' - y = 0, \quad y = Ae^x + Be^{-x}.$$

Boundary condition at $x = 1$ is natural:

$$F_{y'}(1) = 0, \quad F_{y'} = 2y'.$$

So:

$$y'(1) = 0.$$

Apply $y(0) = 0$:

$$A + B = 0 \Rightarrow B = -A.$$

Now compute derivative:

$$y'(x) = Ae^x - Be^{-x} = Ae^x + Ae^{-x}.$$

Thus:

$$y'(1) = A(e + e^{-1}) = 0 \Rightarrow A = 0.$$

Hence $B = 0$ and

$$y(x) = 0.$$

Problem 3

We have

$$J[y] = \int_{x_1}^{x_2} \frac{\sqrt{1+(y')^2}}{y} dx, \quad F(y, y') = \frac{\sqrt{1+(y')^2}}{y}.$$

Since F does not depend on x , we use the Beltrami identity

$$F - y'F_{y'} = C.$$

Compute

$$F_{y'} = \frac{y'}{y\sqrt{1+(y')^2}},$$

so

$$F - y'F_{y'} = \frac{\sqrt{1+(y')^2}}{y} - \frac{(y')^2}{y\sqrt{1+(y')^2}} = \frac{1}{y\sqrt{1+(y')^2}} = C.$$

Hence

$$y\sqrt{1+(y')^2} = \frac{1}{C} = k,$$

so

$$1 + (y')^2 = \frac{k^2}{y^2}, \quad (y')^2 = \frac{k^2 - y^2}{y^2}, \quad \frac{dx}{dy} = \pm \frac{y}{\sqrt{k^2 - y^2}}.$$

Integrating,

$$x = \pm \int \frac{y}{\sqrt{k^2 - y^2}} dy + K = \mp \sqrt{k^2 - y^2} + K.$$

Rearranging gives

$$(x - K)^2 + y^2 = k^2.$$

Therefore the stationary curves are circles

$$(x - K)^2 + y^2 = R^2,$$

with $R^2 = k^2$ and $y > 0$.

Problem 4

Find the stationary function of

$$\int_0^4 [xy' - (y')^2] dx, \quad y(0) = 0, \quad y(4) = 3.$$

Here

$$F = xy' - (y')^2.$$

Compute derivatives:

$$F_y = 0, \quad F_{y'} = x - 2y'.$$

Euler Lagrange:

$$0 - \frac{d}{dx}(x - 2y') = 0.$$

Differentiate:

$$-(1 - 2y'') = 0 \Rightarrow y'' = \frac{1}{2}.$$

Integrate:

$$y' = \frac{1}{2}x + C_1, \quad y = \frac{1}{4}x^2 + C_1x + C_2.$$

Apply $y(0) = 0$: $C_2 = 0$. Apply $y(4) = 3$:

$$3 = \frac{1}{4}(16) + 4C_1 = 4 + 4C_1.$$

$$C_1 = -\frac{1}{4}.$$

Final:

$$y(x) = \frac{1}{4}x^2 - \frac{1}{4}x.$$

Problem 5(a)

Functional:

$$I(y) = \int_0^1 \left[\frac{1}{2}(y')^2 - \frac{1}{4}(1 - y^2)^2 \right] dx, \quad y(0) = 0, \quad y(1) = 1.$$

$$F = \frac{1}{2}(y')^2 - \frac{1}{4}(1 - y^2)^2.$$

Compute:

$$F_y = y(1 - y^2), \quad F_{y'} = y'.$$

Euler Lagrange:

$$y(1 - y^2) - y'' = 0.$$

First integral: multiply by y' .

$$y'(y - y^3) - y''y' = 0.$$

Integrate:

$$\frac{1}{2}(y')^2 - \frac{1}{4}(1 - y^2)^2 = C.$$

This is the first integral.

Problem 5(b)

Constraint:

$$J(y) = \int_0^1 y^2 dx = 1.$$

Introduce a multiplier λ :

$$F = \frac{1}{2}(y')^2 - \frac{1}{4}(1 - y^2)^2 + \lambda y^2.$$

Compute:

$$F_y = y(1 - y^2) + 2\lambda y, \quad F_{y'} = y'.$$

Euler Lagrange:

$$y(1 - y^2) + 2\lambda y - y'' = 0.$$

Final ODE:

$$y'' = y(1 - y^2 + 2\lambda),$$

with boundary conditions

$$y(0) = 0, \quad y(1) = 1.$$